

Quant II

Lab 6

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Today's Agenda

- DID: 2x2, Multi-period single-event, staggered treatment
- `panelView` for visualization
- `HonestDiD`
- Goodman-Bacon decomposition `bacondecomp`
- dCDH `TwoWayFEWeights`
- Sun and Abraham in `fixest`
- Callaway and Sant'Anna estimator & package `did`
-

Parallel Trend Assumptions

Parallel Trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid T_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid T_i = 0]$$

The average *trend in untreated potential outcomes* is the same for treated and control groups.

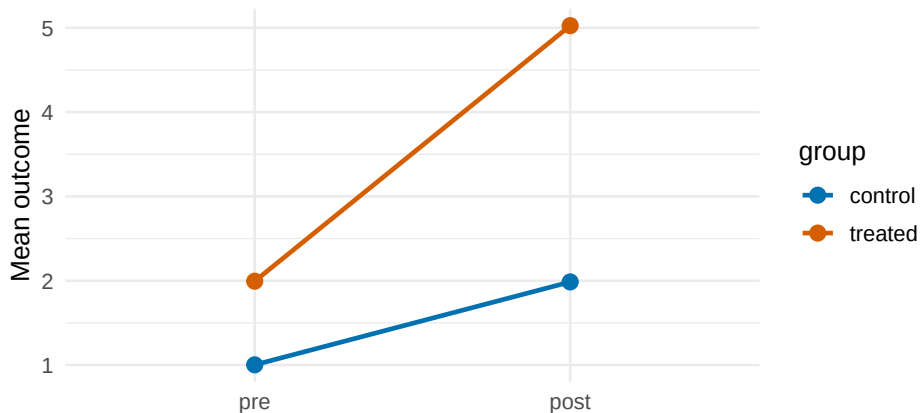
No Anticipation

$$Y_{i0}(1) = Y_{i0}(0) \quad \text{for all } i$$

Treatment has no causal effect prior to implementation.

Simplest setting - 2x2

2x2 DID: mean outcomes by group and period



2x2 DID Estimate

```
# By hand
sim_2x2 |>
  group_by(group, period) |>
  summarise(mean_y = mean(Y), .groups = "drop") |>
  tidyr::pivot_wider(names_from = c(group, period), values_from = mean_y) |>
  mutate(did = (treated_post - treated_pre) - (control_post - control_pre))
```

```
# A tibble: 1 x 5
```

	control_post	control_pre	treated_post	treated_pre	did
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1.99	1.00	5.03	1.99	2.05

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```

```
# A tibble: 1 x 5
```

	control_post	control_pre	treated_post	treated_pre	did
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1.99	1.00	5.03	1.99	2.05

```
# TWFE regression
```

```
fit_twfe <- feols(Y ~ D | id + period, data = sim_2x2)
```

```
# interaction regression
```

```
fit_interaction <- feols(Y ~ treated_group * post, data = sim_2x2)
```

```
# First difference
```

```
sim_diff <- sim_2x2 |>
```

```
  select(id, treated_group, period, Y) |>
```

```
  pivot_wider(names_from = period, values_from = Y) |>
```

```
  mutate(
```

```
    dY = post - pre,
```

```
    treated_fd = treated_group)
```

```
fit_fd <- feols(dY ~ treated_fd, data = sim_diff)
```

2x2 DID

	Equivalent DID estimators		
	TWFE	Interaction	First difference
DID effect	2.048*** (0.062)	2.048*** (0.063)	2.048*** (0.062)
Num.Obs.	4000	4000	2000

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

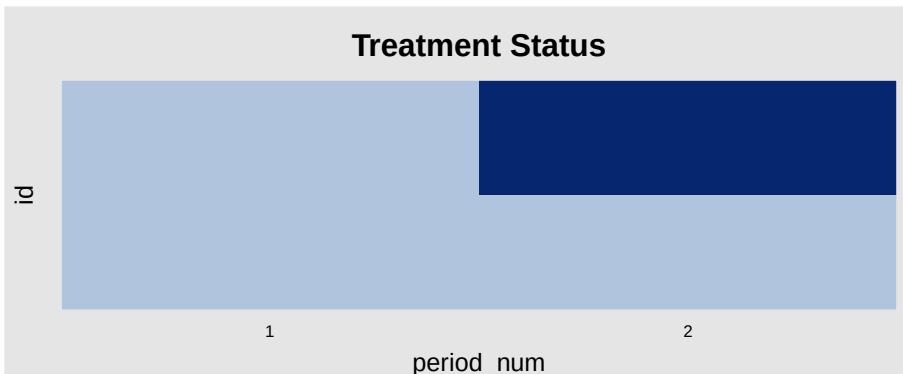
PanelView visualizes

- treatment timing
- whether adoption is one-shot or staggered
- whether there are reversals
- missing data patterns
- whether outcomes move differently across treatment cohorts

PanelView: 2x2 simulated data

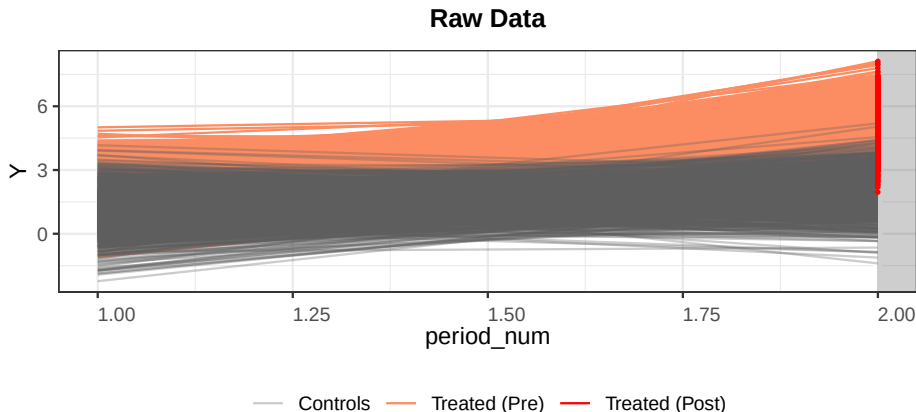
```
sim_2x2_panel <- sim_2x2 |>
  mutate(period_num = if_else(period == "pre", 1L, 2L))

panelview(
  Y = "Y", D = "D", data = sim_2x2_panel,
  index = c("id", "period_num"),
  by.timing = TRUE, display.all = TRUE
)
```



PanelView: 2x2 simulated data

```
panelview(  
  Y = "Y", D = "D", data = sim_2x2_panel, index = c("id", "period_num"),  
  type = "outcome", by.timing = TRUE, display.all = TRUE  
)
```



Multi-period single-event DID

- Multiple pre-treatment periods and/or multiple post-treatment periods
- All treated units receive treatment at the same time

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- We can study **Dynamic ATT** now:

$$\tau_k = \mathbb{E}[Y_{i,k}(1) - Y_{i,k}(0) \mid T_i = 1], \quad k = 1, \dots, K.$$

Multi-period single-event DID

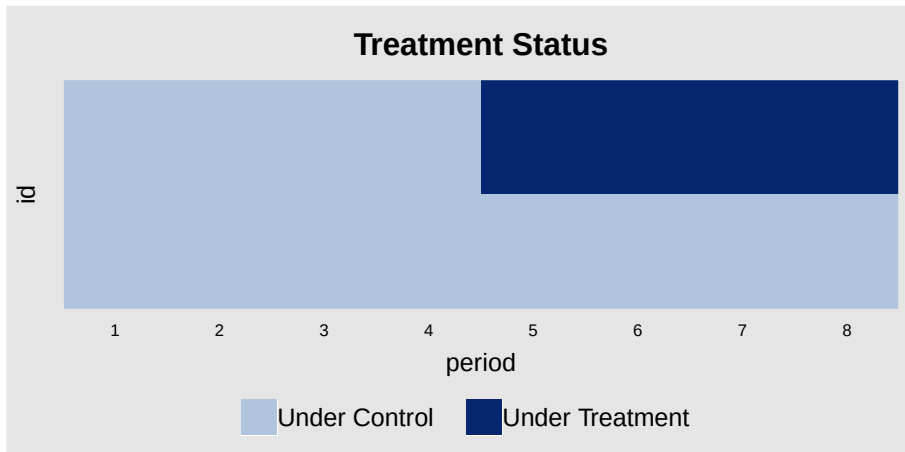
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- Pre-trends and sensitivity analysis

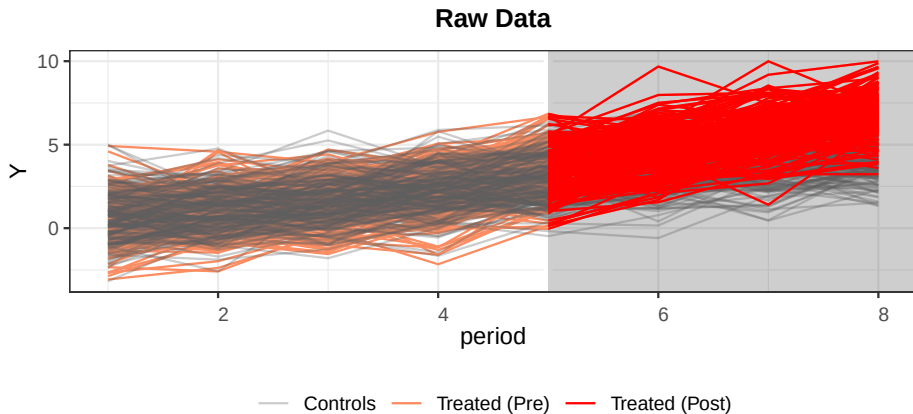
PanelView treatment status

```
panelview(  
  Y = "Y", D = "D", data = single_event,  
  index = c("id", "period"), by.timing = TRUE, display.all = TRUE  
)
```



PanelView outcome

```
panelview(  
  Y = "Y", D = "D", data = single_event, type = "outcome",  
  index = c("id", "period"), by.timing = TRUE, display.all = TRUE  
)
```



Pooled TWFE

```
pooled_twfe <- feols(Y ~ D | id + period, data = single_event,  
                    cluster = ~id)  
etable(pooled_twfe)
```

```

                                pooled_twfe
Dependent Var.:                  Y

D                                1.611*** (0.0709)
Fixed-Effects:  -----
id                                Yes
period                             Yes
-----
S.E.: Clustered                  by: id
Observations                     3,200
R2                               0.82750
Within R2                       0.15278
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Fine in a single-event setting if we have homogeneous effects
- If effects change overtime, pooled TWFE is a weighted average of those effects

Event study specification

Period-specific effects relative to a omitted reference period

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{k=-J \\ k \neq \text{ref}}}^K \delta_k \cdot T_i \times \mathbf{1}(t = k) + \epsilon_{it}$$

- δ_k for $k > 0$: dynamic treatment effects (relative to ref. period).
- δ_k for $k < 0$: pre-treatment lead coefficients
- pre-trends or anticipation

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- δ_k for $k > 0$: dynamic treatment effects (relative to ref. period).
- δ_k for $k < 0$: pre-treatment lead coefficients
- pre-trends or anticipation
- Common choice for reference period
 - last pre-treatment period
 - period before the last pre-treatment period

Event study specification

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- δ_k for $k > 0$: dynamic treatment effects (relative to ref. period).
- δ_k for $k < 0$: pre-treatment lead coefficients
- pre-trends or anticipation
- Common choice for reference period
 - last pre-treatment period
 - period before the last pre-treatment period
 - Shifts what the post-treatment coefficients measure

TWFE event study

```
es_twfe <- feols(  
  Y ~ i(time_to_treat, treated, ref = -1) | id + period,  
  data = single_event, cluster = ~id)  
summary(es_twfe)
```

OLS estimation, Dep. Var.: Y

Observations: 3,200

Fixed-effects: id: 400, period: 8

Standard-errors: Clustered (id)

	Estimate	Std. Error	t value	Pr(> t)
time_to_treat::-4:treated	-0.152249	0.140382	-1.084534	2.7878e-01
time_to_treat::-3:treated	-0.115639	0.137438	-0.841394	4.0063e-01
time_to_treat::-2:treated	0.083140	0.140456	0.591930	5.5423e-01
time_to_treat::0:treated	0.687739	0.143968	4.777028	2.5037e-06 ***
time_to_treat::1:treated	1.182739	0.138704	8.527084	3.1417e-16 ***
time_to_treat::2:treated	1.921987	0.144621	13.289797	< 2.2e-16 ***
time_to_treat::3:treated	2.467795	0.137403	17.960242	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.916857 Adj. R2: 0.814956

Within R2: 0.208483

TWFE event study

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  Y ~ i(time_to_treat, treated, ref = -1) | id + period,  
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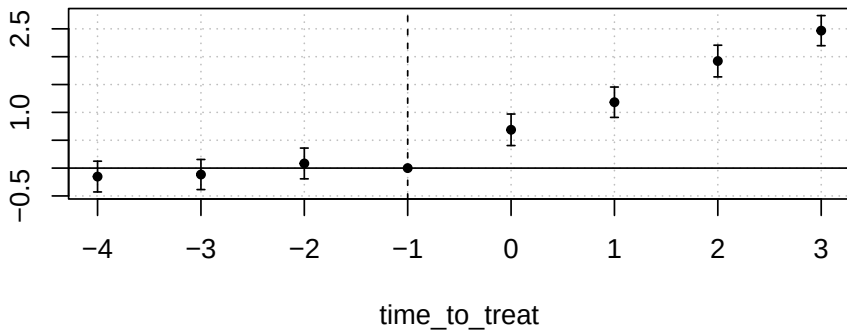
```
coef_es <- coef(es_twfe)  
post_coefs <- coef_es[grepl("time_to_treat::[0-9]+:treated", names(coef_es))]  
mean(post_coefs)
```

[1] 1.565065

TWFE event study

```
ipplot(es_twfe, main = "")
```

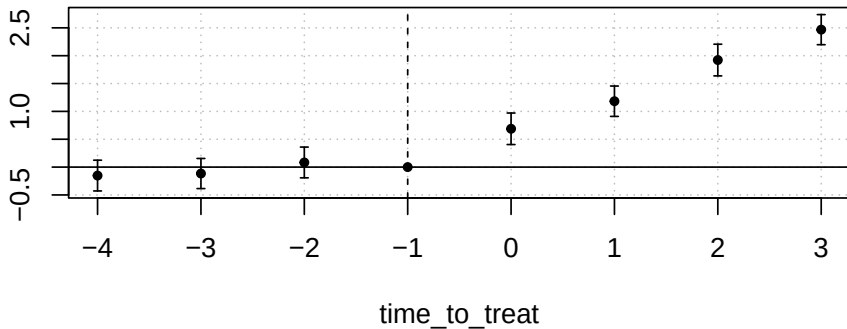
Estimate and 95% Conf. Int.



TWFE event study

```
iplot(es_twfe, main = "")
```

Estimate and 95% Conf. Int.

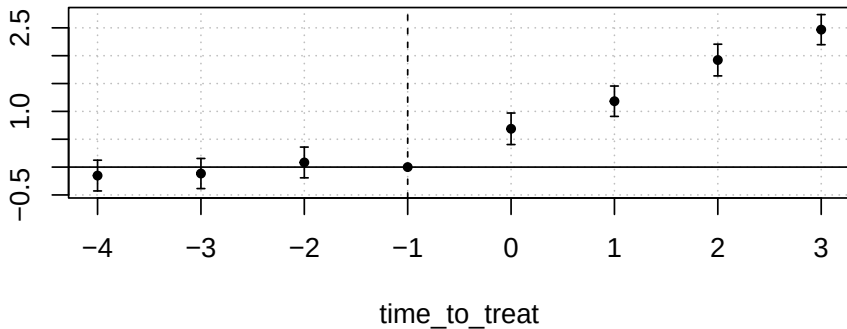


- Under parallel trends and no anticipation, pre-treatment lead coefficients should be close to zero

TWFE event study

```
iplot(es_twfe, main = "")
```

Estimate and 95% Conf. Int.

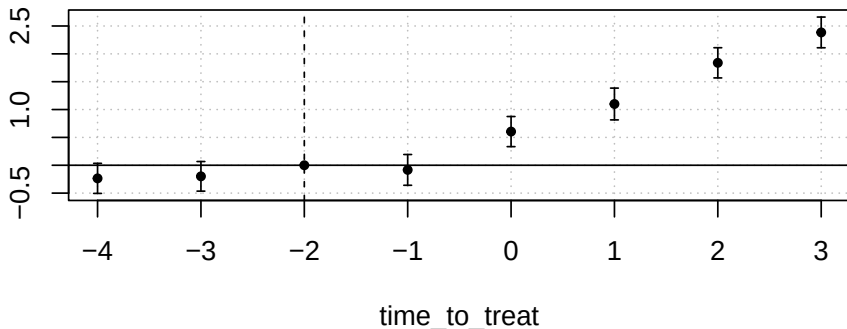


- Under parallel trends and no anticipation, pre-treatment lead coefficients should be close to zero
- But a flat pre-trend is not proof

TWFE event study - no anticipation

```
es_twfe_ref2 <- feols(  
  Y ~ i(time_to_treat, treated, ref = -2) | id + period,  
  data = single_event, cluster = ~id  
)  
iplot(es_twfe_ref2, main = "")
```

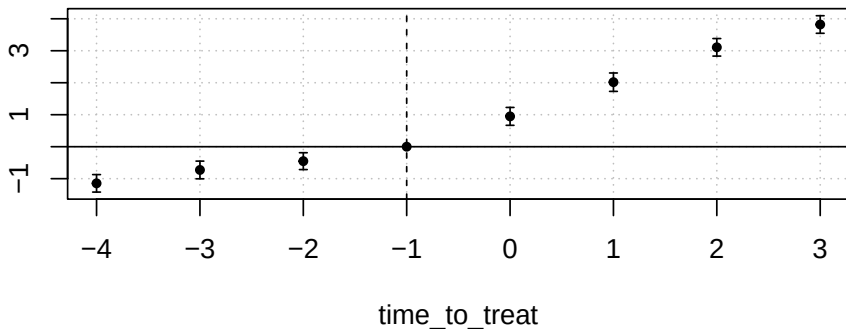
Estimate and 95% Conf. Int.



Pre-trend

Estimate and 95% Conf. Int.

Effect on Y



If post-treatment trend violations were no larger than some multiple of the pre-treatment pattern, would the treatment effect still be statistically distinguishable from zero?

If post-treatment trend violations were no larger than some multiple of the pre-treatment pattern, would the treatment effect still be statistically distinguishable from zero?

- Estimate an event study
- Use the pre-treatment coefficients to inform possible violations after the treatment
- A sensitivity parameter $\bar{M}$
- Check how the confidence set changes as $\bar{M}$ increases

HonestDiD - Single Event

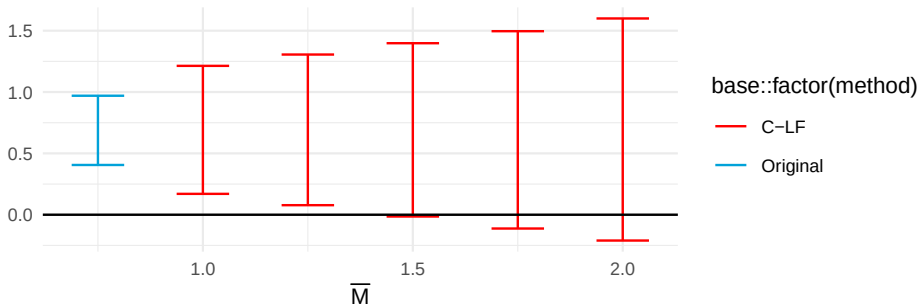
```
betahat <- summary(es_twfe)$coefficients
sigma <- summary(es_twfe)$cov.scaled
delta_rm_results <- HonestDiD::createSensitivityResults_relativeMagnitudes(
  betahat = betahat, #coefficients
  sigma = sigma, #covariance matrix
  numPrePeriods = 3, #num. of pre-treatment coefs
  numPostPeriods = 4, #num. of post-treatment coefs
  Mbarvec = seq(1,2,by=0.25) #values of Mbar
)
delta_rm_results
```

A tibble: 5 x 5

	lb	ub	method	Delta	Mbar
	<dbl>	<dbl>	<chr>	<chr>	<dbl>
1	0.170	1.21	C-LF	DeltaRM	1
2	0.0778	1.31	C-LF	DeltaRM	1.25
3	-0.0144	1.40	C-LF	DeltaRM	1.5
4	-0.112	1.50	C-LF	DeltaRM	1.75
5	-0.210	1.60	C-LF	DeltaRM	2

HonestDiD - Single Event

```
originalResults <- HonestDiD::constructOriginalCS(  
  betahat = betahat, sigma = sigma,  
  numPrePeriods = 3, numPostPeriods = 4)  
HonestDiD::createSensitivityPlot_relativeMagnitudes(delta_rm_results, originalResul
```



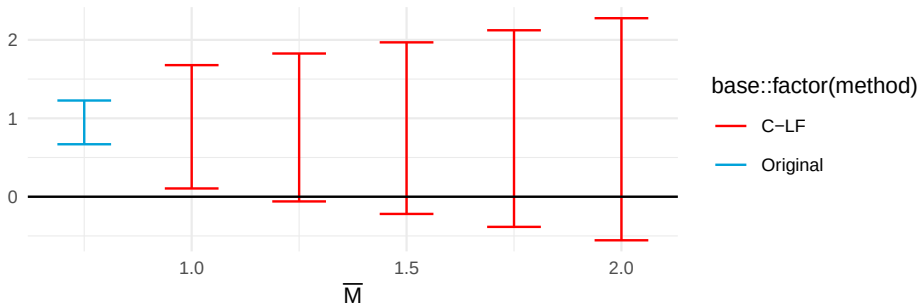
HonestDiD - Single Event Example 2

```
betahat_pt <- summary(es_twfe_pretrend)$coefficients  
sigma_pt <- summary(es_twfe_pretrend)$cov.scaled
```

```
delta_rm_results <- HonestDiD::createSensitivityResults_relativeMagnitudes(  
  betahat = betahat_pt, #coefficients  
  sigma = sigma_pt, #covariance matrix  
  numPrePeriods = 3, #num. of pre-treatment coeffs  
  numPostPeriods = 4, #num. of post-treatment coeffs  
  Mbarvec = seq(1,2,by=0.25) #values of Mbar )  
)
```

```
originalResults <- HonestDiD::constructOriginalCS(  
  betahat = betahat_pt, sigma = sigma_pt,  
  numPrePeriods = 3, numPostPeriods = 4)
```

```
HonestDiD::createSensitivityPlot_relativeMagnitudes(delta_rm_results, originalResults) + theme_minimal(base_size = 12)
```



Staggered Adoption - common in applied works

- Different units adopt in different periods
- Dynamic treatment effects are allowed

Staggered Adoption - common in applied works

- Different units adopt in different periods
- Dynamic treatment effects are allowed

Under staggered adoption, the TWFE coefficient is a weighted average of many 2x2 comparisons.

Some are clean:

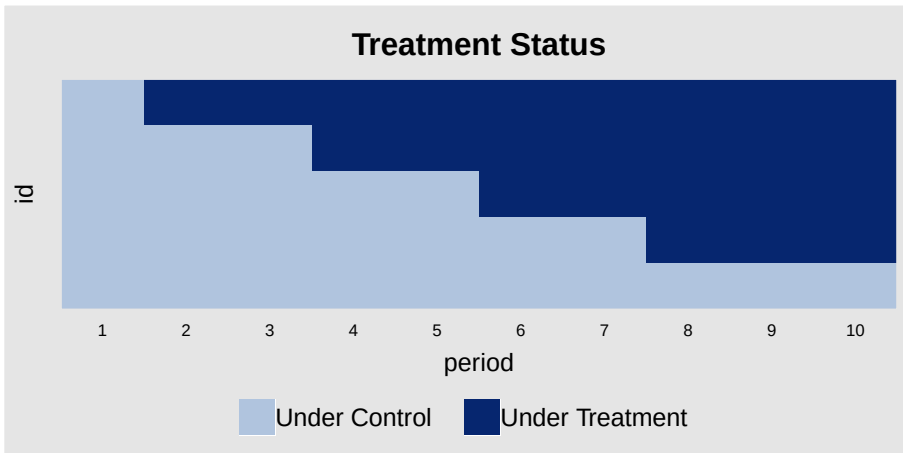
- newly treated versus never treated
- earlier treated versus later treated while the later cohort is still untreated

Some are bad:

- later treated units compared to already treated units

Simulated Example PanelView

```
panelview(  
  Y = "y", D = "treated", data = sim_staggered,  
  index = c("id", "period"), by.timing = TRUE, display.all = TRUE)
```



Pooled TWFE Estimate

```
twfe_fit <- feols(y ~ treated | id + period, data = sim_staggered, cluster = ~id)
etable(twfe_fit)
```

```

                                twfe_fit
Dependent Var.:                  y

treated                1.683*** (0.0466)
Fixed-Effects:  -----
id                      Yes
period                 Yes
-----
S.E.: Clustered          by: id
Observations             5,000
R2                       0.79110
Within R2                0.24607
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Goodman-Bacon Decomposition

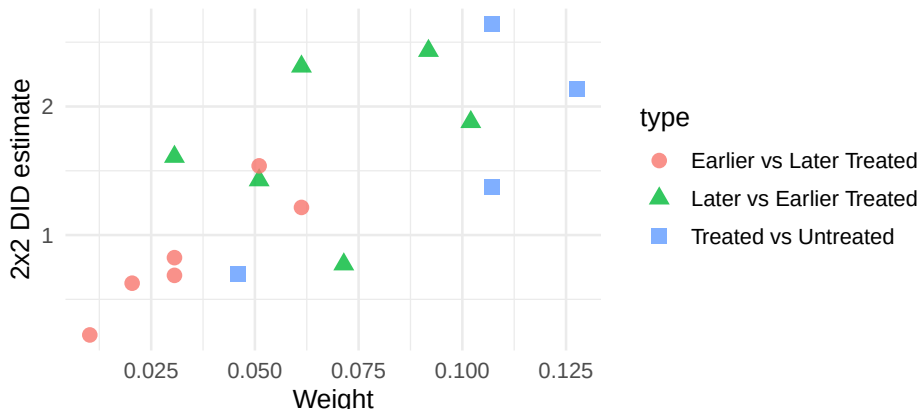
```
library(bacondecomp)
bacon_out <- bacon(
  y ~ treated, data = sim_staggered, id_var = "id", time_var = "period")
```

	type	weight	avg_est
1	Earlier vs Later Treated	0.20408	1.04943
2	Later vs Earlier Treated	0.40816	1.79806
3	Treated vs Untreated	0.38776	1.89574

Goodman-Bacon Decomposition

```
ggplot(bacon_out, aes(x = weight, y = estimate, color = type, shape = type)) +  
  geom_point(size = 3, alpha = 0.8) +  
  labs(title = "Goodman-Bacon decomposition",  
        x = "Weight", y = "2x2 DID estimate") +  
  theme_minimal(base_size = 12)
```

Goodman-Bacon decomposition



dCDH TwoWayFEWeights

```
library(TwoWayFEWeights)
twfe_wts <- twowayfeweights(data = sim_staggered, Y = "y", G = "id",
                           T = "period", D = "treated", type = "feTR", summary_measures = "weights")
twfe_wts
```

Under the common trends assumption, the TWFE coefficient β , equal to 1.6832, estimates a weighted sum of 2400 ATTs. 1600 ATTs receive a positive weight, and 800 receive a negative weight.

Treat. var: treated	ATTs	Σ weights

Positive weights	1600	1.1939
Negative weights	800	-0.1939

Total	2400	1

Summary Measures:

TWFE Coefficient (β): 1.6832

min (Δ) compatible with β and $\Delta_{TR} = 0$: 1.3246

min (Δ) compatible with treatment effect of opposite sign than β in all (g,t)

Reference: Corollary 1, de Chaisemartin, G and D'Haultfoeuille, X (2020a)

Callaway & Sant'Anna

- Default control group is “nevertreated”, can be set to “notyettreated”
- default estimation method is doubly robust from DRDID

```
library(did)
cs_dr <- att_gt(yname = "y",      # outcome variable
               gname = "g",      # group variable
               idname = "id",    # unit identifier
               tname = "period", # year variable
               xformula = ~1,    # covariates (if any)
               data = sim_staggered,
               est_method = "dr",
               control_group = "nevertreated"
)
```

Callaway & Sant'Anna

Aggregated overall ATT:

```
aggte(cs_dr, type = "simple", na.rm=T)
```

Call:

```
aggte(MP = cs_dr, type = "simple", na.rm = T)
```

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences

ATT	Std. Error	[95% Conf. Int.]
1.468	0.0721	1.3266 1.6093 *

Signif. codes: `*' confidence band does not cover 0

Control Group: Never Treated, Anticipation Periods: 0

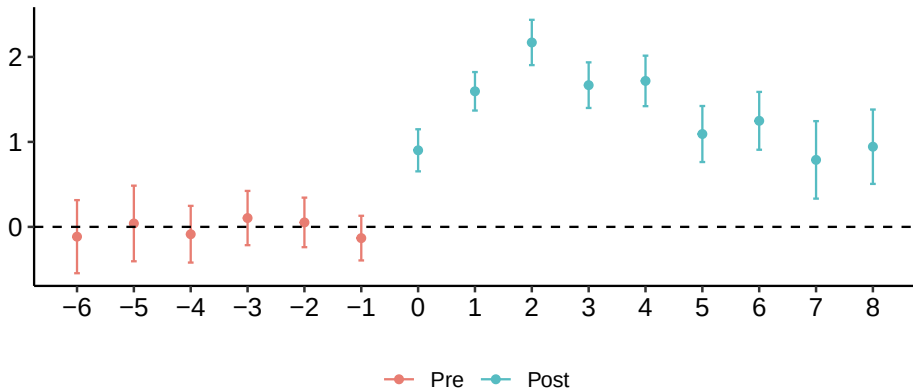
Estimation Method: Doubly Robust

Callaway & Sant'Anna

Dynamic ATT (Event-Study):

```
cs_dynamic <- aggte(cs_dr, type = "dynamic")  
ggdid(cs_dynamic)
```

Average Effect by Length of Exposure



Sun-Abraham with fixest

Aggregated overall ATT:

```
sa_fit <- feols(y ~ sunab(g_sa,period) | id + period,  
               data = sim_staggered, cluster = ~id)  
summary(sa_fit, agg = "ATT")
```

OLS estimation, Dep. Var.: y

Observations: 5,000

Fixed-effects: id: 500, period: 10

Standard-errors: Clustered (id)

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

ATT	1.46795	0.06425	22.8475	< 2.2e-16 ***
-----	---------	---------	---------	---------------

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.752169 Adj. R2: 0.805123

Within R2: 0.373213

Sun-Abraham with fixest

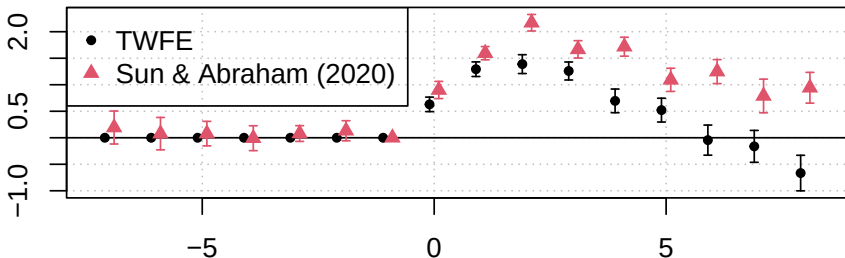
Event study plot:

```
# compare to twfe event study
twfe_es_fit <- feols(y ~ i(time_to_treat, treated, ref = -1) | id + period,
  data = sim_staggered, cluster = ~id)

iplot(list(twfe_es_fit, sa_fit)); legend("topleft", col = c(1,2), pch = c(20, 17),
  legend = c("TWFE", "Sun & Abraham (2020)"))
```

Estimate and 95% Conf. Int.

Effect on y

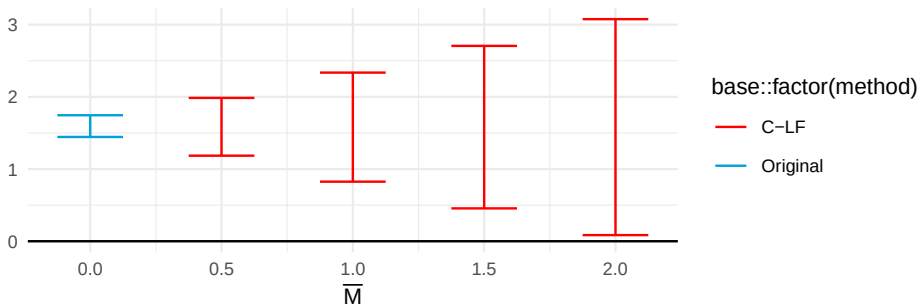


HonestDiD with staggered treatment

```
source("honest-helpers.R"))
# Need to specify the "universal" type base period to use honestDiD
es_out <- att_gt(yname = "y", gname = "g", idname = "id",
                 tname = "period", xformula = ~1, control_group = "nevertreated",
                 data = sim_staggered, est_method = "reg",
                 base_period = "universal")
es <- aggte(es_out, type = "dynamic")
sensitivity_results <- honest_did.AGGTEobj(
  es, e = 1, type = "relative_magnitude", Mbarvec = seq(0.5, 2, by = 0.5),
  grid.ub = 10, grid.lb = -10, gridPoints = 2000)
```

Honest DiD

```
HonestDiD::createSensitivityPlot_relativeMagnitudes(sensitivity_results$robust_ci,  
sensitivity_results$orig_ci)+theme_minimal(base_size = 10)
```



Other resources

- PanelMatch for matching
- DIDmultiplegt de Chaisemartin & D'Haultfœuille DID methods.
- Yiqing Xu's Tutorials on YouTube
- Sant'Anna's Lectures on DiD

Example Study

Mafia and political quality: Daniele & Geys (EJ 2015)

ORGANISED CRIME, INSTITUTIONS AND POLITICAL QUALITY: EMPIRICAL EVIDENCE FROM ITALIAN MUNICIPALITIES*

Gianmarco Daniele and Benny Geys

This article assesses how legal institutions affect the influence of politically active criminal organisations on the human capital of elected politicians using data from over 1,500 Southern Italian municipalities in the period 1985–2011. It exploits municipal government dissolutions imposed by the national government for (presumed) mafia infiltration as a source of exogenous variation in the presence of politically active criminal organisations. The results support theoretical predictions that the average education level of local politicians significantly increases when active mafia infiltration of local politics is remedied through the implementation of a stricter legal-institutional framework.

Example Study

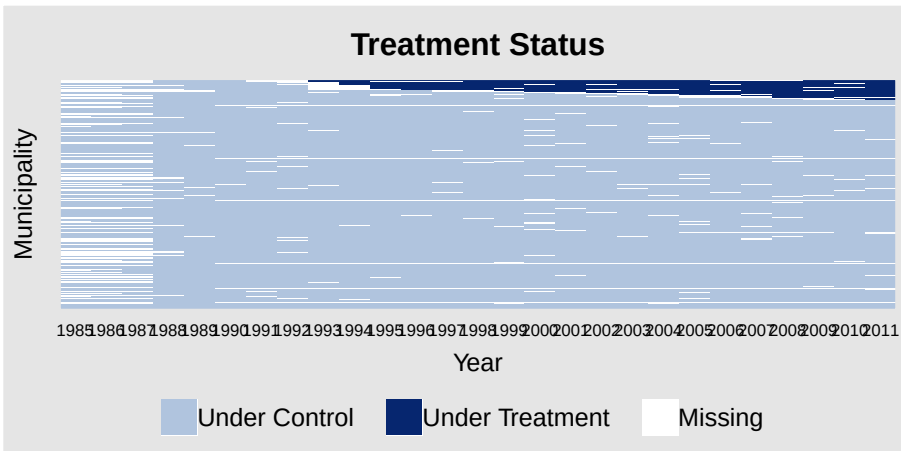
- What is the effect of removing mafia connections on the quality of politicians?
- Staggered dissolution of municipal governments in Italian south after criminal investigations
- Outcome: education of government members (average)

Example Study

- What is the effect of removing mafia connections on the quality of politicians?
- Staggered dissolution of municipal governments in Italian south after criminal investigations
- Outcome: education of government members (average)
- “Unlike the traditional DiD model, which relies on a shock at one point in time across all treated jurisdictions, we can exploit the fact that dissolutions did not take place at the same point in time for each municipality [. . .] to further strengthen our identification’ ’

panelView treatment status

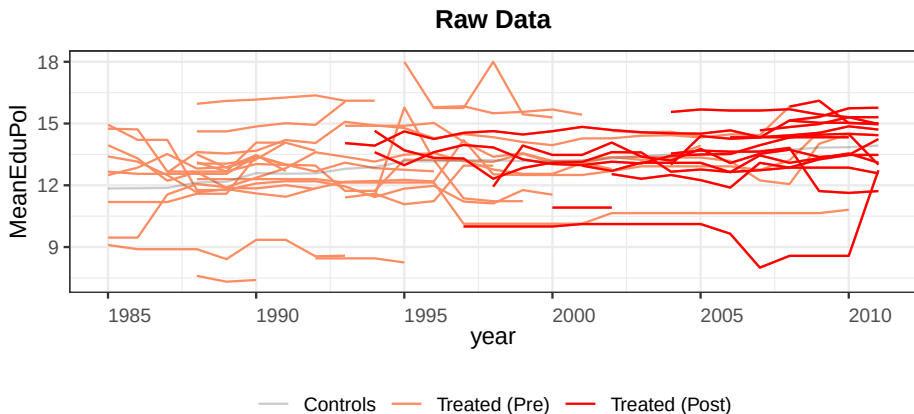
```
panelview(  
  MeanEduPol ~ becomgeneral_recode, data = mafia_df,  
  index = c("ID_municip", "year"), xlab = "Year", ylab = "Municipality",  
  display.all = TRUE, gridOff = TRUE, by.timing = TRUE)
```



panelView: outcomes by treatment cohort

```
panelview(  
  MeanEduPol ~ befcomgeneral_recode, data = mafia_df, index = c("ID_municip", "year"  
  by.timing = TRUE, display.all = FALSE, type = "outcome", by.cohort = TRUE)
```

Number of unique treatment histories: 18



Pooled TWFE

```
# each municipality have its own linear time trend.
mafia_twfe <- feols(
  MeanEduPol ~ becomgeneral_recode | ID_municip[trend] + year,
  data = mafia_df, cluster = ~ID_municip)
etable(mafia_twfe)
```

```

                                mafia_twfe
Dependent Var.:                MeanEduPol

becomgeneral_recode 0.3213* (0.1249)
Fixed-Effects:      -----
ID_municip          Yes
year                Yes
Varying Slopes:     -----
trend (ID_municip)  Yes

-----
S.E.: Clustered      by: ID_municip
Observations         39,191
R2                   0.78400
Within R2            0.00082
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Data Prep

Define treatment groups based on their first year of treatment

```
mafia_df <- mafia_df %>% group_by(desc_comune) %>%  
  mutate(group = case_when(mafiaben==1 ~ min(year[befcomgenera  
                           mafiaben==0 ~ 0)) %>%  
  ungroup()
```

Callaway & Sant'Anna

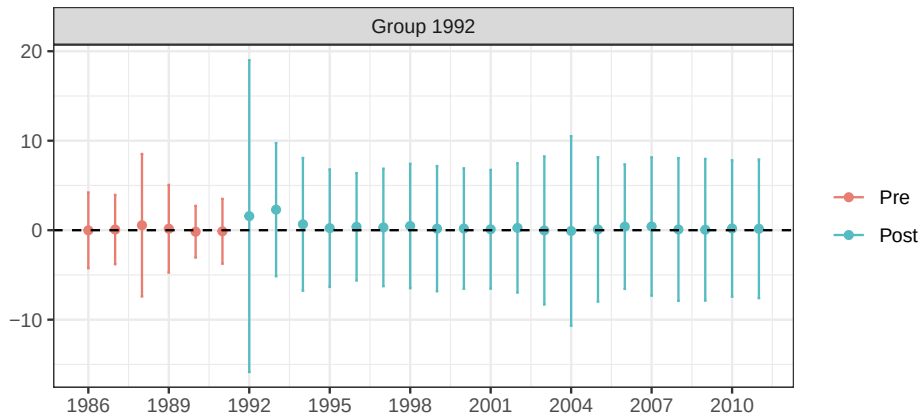
```
library(did)
set.seed(1)
cs_dr <- att_gt(yname = "MeanEduPol",      # outcome variable
               gname = "group",            # group variable
               idname = "ID_municip",      # unit identifier
               tname = "year",             # year variable
               xformla = ~1,               # covariates (if any)
               data = mafia_df,            # data
               est_method = "dr",          # estimation method
               allow_unbalanced_panel = T, # allow unbalanced panel
               control_group = "nevertreated" # control group
)
tidy(cs_dr)
```

	term	group	time	estimate	std.error	conf.low	conf.high
1	ATT(1991,1986)	1991	1986	0.132434849	0.23366798	-5.02137268	5.2862424
2	ATT(1991,1987)	1991	1987	0.439155108	0.49691475	-10.52085304	11.3991633
3	ATT(1991,1988)	1991	1988	-0.904125651	0.49434466	-11.80744771	9.9991964
4	ATT(1991,1989)	1991	1989	-0.073122138	0.16025542	-3.60773388	3.4614896
5	ATT(1991,1990)	1991	1990	-0.341371843	0.13043611	-3.21828542	2.5355417
6	ATT(1991,1991)	1991	1991	NA	NA	NA	NA
7	ATT(1991,1992)	1991	1992	NA	NA	NA	NA
8	ATT(1991,1993)	1991	1993	0.716874927	0.28768560	-5.62835137	7.0621012
9	ATT(1991,1994)	1991	1994	0.144372669	0.29051864	-6.26333947	6.5520848
10	ATT(1991,1995)	1991	1995	0.042224356	0.28301863	-6.20006667	6.2845154
11	ATT(1991,1996)	1991	1996	-0.105071221	0.31279308	-7.00407085	6.7939284
12	ATT(1991,1997)	1991	1997	0.074856599	0.36569808	-7.99102183	8.1407350
13	ATT(1991,1998)	1991	1998	0.452671768	0.28707131	-5.87900573	6.7843493
14	ATT(1991,1999)	1991	1999	0.296876207	0.30145885	-6.35213448	6.9458869
15	ATT(1991,2000)	1991	2000	0.251675147	0.25333861	-5.33599002	5.8393403
16	ATT(1991,2001)	1991	2001	0.488689856	0.32926428	-6.77360043	7.7509801
17	ATT(1991,2002)	1991	2002	0.258719244	0.37534648	-8.01996535	8.5374038

Callaway & Sant'Anna - visualization

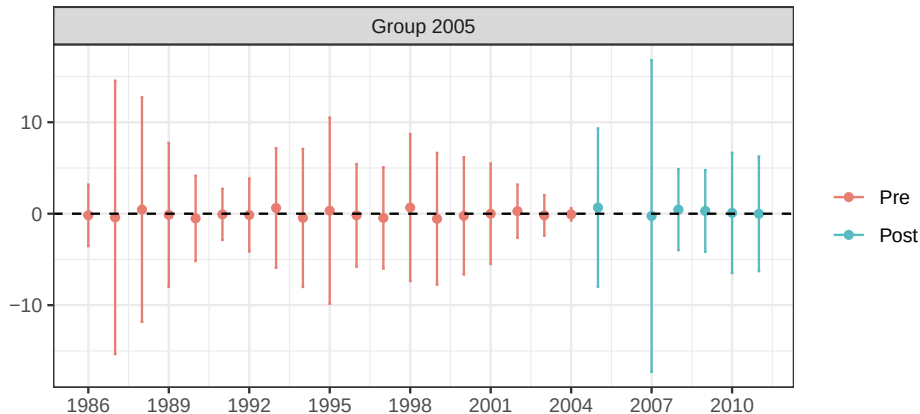
By default, it makes a plot for each group, here we select a few

```
ggdid(cs_dr, group = 1992, xgap=3, title="", theming=F)+ theme_bw()
```



Callaway & Sant'Anna - visualization

```
ggdid(cs_dr, group = 2005, xgap=3, title="", theming=F)+ theme_bw()
```



Aggregate estimates

```
aggte(cs_dr, type = "dynamic", na.rm=T)
```

Call:

```
aggte(MP = cs_dr, type = "dynamic", na.rm = T)
```

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences v

Overall summary of ATT's based on event-study/dynamic aggregation:

ATT	Std. Error	[95% Conf. Int.]
0.2294	0.1186	-0.0031 0.4618

Dynamic Effects:

Event time	Estimate	Std. Error	[95% Simult. Conf. Band]
-24	0.3312	0.5891	-1.3774 2.0397
-23	-0.4090	0.2522	-1.1404 0.3224
-22	0.1171	0.2155	-0.5079 0.7420
-21	-0.0423	0.3258	-0.9872 0.9027
-20	0.2648	0.1763	-0.2465 0.7760
-19	-0.3340	0.1451	-0.7550 0.0870
-18	-0.3085	0.2591	-1.0600 0.4430
-17	0.1776	0.2212	-0.2102 0.4640

TWFE Weights

```
wgt_diag <- twowayfeweights(  
  mafia_df, "MeanEduPol", "ID_municip", "year",  
  "befcomgeneral_recode", type = "feTR", summary_measures = TRUE)  
wgt_diag
```

Create Staggered Variables

```
cohorts <- mafia_df |>
  group_by(ID_municip) |>
  summarise(
    first_treat = if (any(befcomgeneral_recode == 1))
      min(year[befcomgeneral_recode == 1]) else 0L,
    ever_treated = as.integer(first_treat > 0),
    .groups = "drop")

mafia_staggered <- mafia_df |>
  left_join(cohorts, by = "ID_municip") |>
  mutate(
    time_to_treat = if_else(ever_treated == 1, year - first_treat, 0L))
```

Create Staggered Variables

```
mafia_staggered |>
  select(desc_comune, year, becomgeneral_recode, first_treat, ever_treated, time_to_treat) |>
  arrange(desc_comune, year) |>
  filter(desc_comune == "ACERRA") |>
  head(15)
```

A tibble: 15 x 6

	desc_comune	year	becomgeneral_recode	first_treat	ever_treated	time_to_treat
	<chr>	<int>	<dbl>	<int>	<int>	<int>
1	ACERRA	1985	0	1993	1	-8
2	ACERRA	1986	0	1993	1	-7
3	ACERRA	1987	0	1993	1	-6
4	ACERRA	1988	0	1993	1	-5
5	ACERRA	1989	0	1993	1	-4
6	ACERRA	1990	0	1993	1	-3
7	ACERRA	1991	0	1993	1	-2
8	ACERRA	1992	0	1993	1	-1
9	ACERRA	1993	1	1993	1	0
10	ACERRA	1994	1	1993	1	1
11	ACERRA	1995	1	1993	1	2
12	ACERRA	1996	1	1993	1	3
13	ACERRA	1997	1	1993	1	4
14	ACERRA	1998	1	1993	1	5
15	ACERRA	1999	1	1993	1	6

TWFE Event Study & Sun-Abraham with fixest

```
twfe_es_fit <- feols(  
  MeanEduPol ~ i(time_to_treat, ever_treated, ref = -1) | ID_municip + year,  
  data = mafia_staggered, cluster = ~ID_municip)  
mafia_sa <- mafia_staggered |>  
  mutate(first_treat_sa = if_else(first_treat == 0, 10000L, first_treat))  
sa_fit <- feols(  
  MeanEduPol ~ sunab(first_treat_sa, year) | ID_municip + year,  
  data = mafia_sa,  
  cluster = ~ID_municip)
```

TWFE Event Study & Sun-Abraham with fixest Plot

